

Development of Blind Football Play-by-play System for Visually Impaired Spectators

Tangible Sports

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ABSTRACT

In the field of sports, visually impaired spectators are unfortunately at a disadvantage in terms of understanding the precise developments of a game. To provide such information to them, making play-by-play announcements of the live coverage of the game is the best approach; however, this is infeasible as it is difficult to employ a professional commentator for every game. Therefore, this study proposes a blind football play-by-play system, combining a tactile graphic display function and a position acquisition function, to aid visually impaired spectators understand game developments. To conduct a basic study, an experimental system was developed herein, which detects the positions of the players and ball using image processing technology and deep learning, and shows these positions via a refreshable tactile display. The validity of the proposed system was then established through subjective experiments. This system provides a rich user experience to the spectators as well as being a powerful feedback tool to the players. Moreover, an audio-tactile graphic system, which can be used to overcome tactile cognitive limitations, can be implemented as an application of the system in the future.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction devices; Haptic devices; Accessibility; Accessibility technologies; Interaction design; Systems and tools for interaction design; • **Information systems** → Information systems applications; Digital libraries and archives.

KEYWORDS

Tactile graphic display, Feedback tool, Support system, User experience

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1 INTRODUCTION

Blind football has become increasingly popular in the field of para sports, with worldwide championships being conducted regularly by the International Blind Sports Federation (IBSA) [1], which are broadcast on the internet and even on TV in Japan. Consequently, the number of the spectators attending these events has also been increasing. These spectators can be categorized into three groups [2]. The first group consists of athletes participating in the game, the second group includes individuals such as family members or friends of the athletes, and the third group of members includes unrelated individuals with disabilities. Particularly, in the case of blind football, several spectators are visually impaired. However, visually impaired spectators face difficulty in understanding the precise developments of these games. To provide such information to them, making play-by-play announcements of the live coverage of the games is the best solution, although this is infeasible as it is difficult to employ a professional commentator for every game. Therefore, Panasonic developed a support system to broadcast live announcements to the smartphones or tablets of visually impaired spectators via network [3]. However, this solution, albeit discreet and effective, requires a professional sports commentator and is difficult to realize in the case of local matches.

Therefore, this study aims to develop a support system for visually impaired spectators, which can automatically provide game updates in real time, even in local blind football matches [4].

2 SYSTEM CONFIGURATION

To develop the blind football play-by-play system for the visually impaired spectators, an input and output method was designed considering the concept of the system, as shown in Figure 1. The input method should be able to correctly detect the positions of the players and ball, and the output method should effectively provide this position information to the spectators. For the output method, a synthesized voice and/or refreshable tactile display are most plausible. To realize automatic live coverage, voice announcements would be the most favored solution, although the tactile graphic display is easily applicable. A consumer version of device by KGS Corp. can be utilized, which is widely employed in Japan, and it has already been established that the blind can understand a significant amount of graphical information from this device [5].

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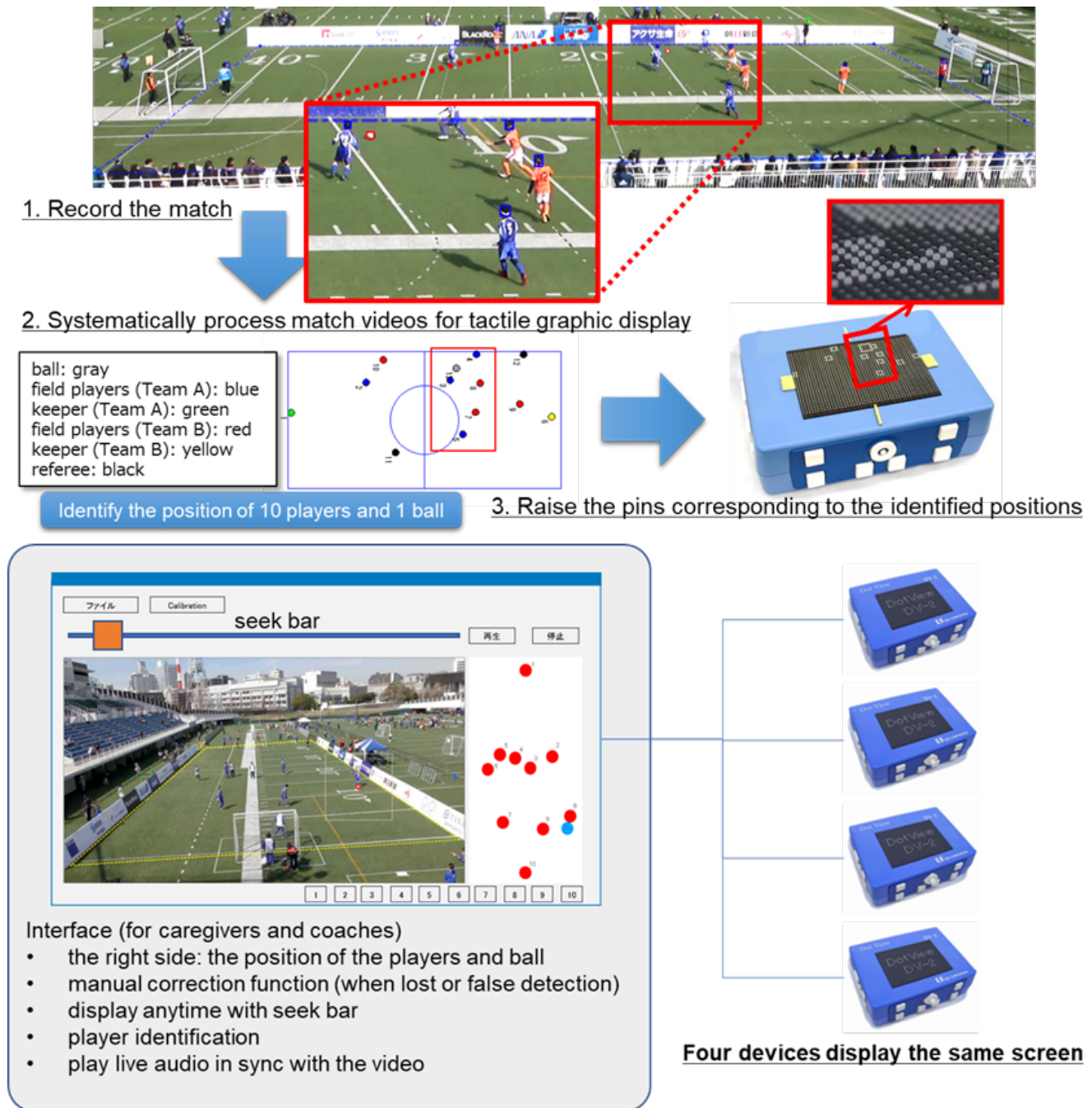


Figure 1: Concept of the proposed system.

The length and width of the soccer field are 40 and 20 m, respectively, while the length and width of the tactile display are 0.4 and 0.2 m (40 × 20 cm). Therefore, if the result of the position measurement is multiplied by 1/100 and the corresponding pin of the tactile display rises, it can be projected as a bird's-eye-view layout of the ball and players on the field. By recognizing the edges of the tactile display as the end lines and side lines, the spectator

can understand the relative positions of the ball and players by touch.

For the input method, the utilization of a camera to detect the position of the players and ball was considered. If the cameras are equipped with an image processing system, the players do not need to wear special devices. This proposed method consists of a position acquisition system, playback system, and tactile graphic

display. The acquired position data of the 10 players and ball are displayed on the tactile graphic display. This system provides a rich user experience to the spectators.

2.1 Tactile graphic display

The majority of tactile graphic displays developed in the past were only capable of unidirectional communication [6–8]. In this study, four bimorph modules (SC–10, provided by KGS Inc.) were used to assemble the tactile display. Each module has a pin configuration of 32×12 , with a 2.4-mm spacing. The pin is 1.3 mm in diameter and must be pushed down with a force of at least 0.098 N to touch the surface. The total tactile surface has a 32×48 configuration that is activated simultaneously. The height of the tactile pin, when activated by a data signal, is 0.7 mm. The refresh time of the entire surface is approximately 50 ms. The device is interfaced with the host PC through USB or serial communication. The tactile graphic display reads the comma-separated values (CSV) file, which saves and displays the position information of the player and ball, as acquired by the position acquisition system.

2.2 Position acquisition system

The position acquisition system automatically tracks the position information of the players and ball, using image processing technology and deep learning. The video data were acquired at the KPMG Cup Blind Soccer Club Team Championship on February 10, 2019, and recorded in 4K at 30 fps, along with a panoramic view of the soccer field, using a professional video camera and not a stereo camera. For annotation, rectangles and labels were added to any object with more than 50,000 points and more than 4,700 frames. This system detects 10 players, 2 referees, and 1 ball from the input image, and converts this data into their position on the soccer field. The obtained position information was saved as a CSV file.

3 EXPERIMENTAL SYSTEM AND RESULT

After designing the input/output methods and devices, a set of experimental systems was established to conduct two basic studies and a verification study of the proposed system.

Figure 2 shows the experimental setup of the following three experiments. The system comprised a graphic tactile display and computer. The participant was made to sit in front of the

tactile graphic display, wearing a protective eye mask and earmuffs. The experimental procedure followed in all three experiments was approved by the Tokyo Metropolitan Industrial Technology Research Institute ethics committee. All participants provided informed prior consent, according to the institutional requirements.

3.1 Experiment 1 - Ability to discriminate between different frequencies

The purpose of this experiment was to identify the combination of the different vibration frequencies that allow the user to easily distinguish between the two teams.

3.1.1 Participants. Seven males (aged 19 to 52 years, 28.6 ± 12.5) volunteered as participants in this study. Table 1 shows the details of the participants. Three of them were blind, and four were visually impaired.

3.1.2 Procedure. The participants subjectively evaluated the ease of differentiation between two vibrations with different vibration frequencies on a five-point scale using the SD method. The vibration frequencies that can be controlled by the tactile graphic display used in this system were in an increasing order of 0, 1, 2, 3, 4, 8, 16, and 32 Hz. As shown in Figure 3, a square was drawn with the pins of 8×8 as one set, designated as 0, 1, 2, and 3 Hz from the upper left to upper right, and 4, 8, 16, and 32 Hz from the lower left to lower right. The process was initiated with a combination of 0 Hz and 1 Hz, followed by a combination of 0 and 2 Hz, then 0 and 3 Hz, and so on up to a combination of 16 and 32 Hz. The process was then initiated with the combination of 32 Hz and 16 Hz, proceeded to 16 Hz and 8 Hz, then 16 Hz and 4 Hz, and so on down to the combination of 1 Hz and 0 Hz. The participants were not provided instructions on the operation.

3.1.3 Results. Table 2 shows the average of the scores of the subjective evaluation. It is observed that 0 Hz and 1 Hz were highly evaluated in combination with any frequency. The participants' evaluations were low for the combinations of adjacent frequencies above 3 Hz. The combination of 8 Hz and 16 Hz and the combination of 16 Hz and 32 Hz had particularly low scores. From these results, 0, 1, 2, 8, and 32 Hz were deemed most suitable for distinguishing two different vibrations.



Figure 2: Experimental output device for all three experiments.

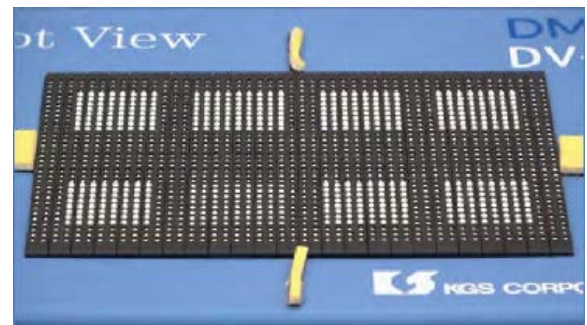


Figure 3: Setup for Experiment 1.

Table 1: Participants in Experiment 1

	Age	Visual acuity (yard-pound system)	Congenital/acquired	Type of characters used	Method to obtain information	Experience of habitually watching sports	Remarks
1	52	under 20/2000	Acquired	Braille	Voice, braille	Yes	
2	43	under 20/2000	Congenital	Braille	Voice, braille	Yes	
3	19	under 20/2000	Congenital	Braille	Voice, braille	Yes	
4	19	under 20/2000	Congenital	Braille	Voice, braille	Yes	
5	20	20/1000 – 20/666	Congenital	Braille, printed	Voice, enlargement, braille	Yes	
6	20	20/1000 – 20/666	Acquired	Printed	Same as sighted person	Yes	Blind soccer player
7	27	20/1000 – 20/666	Acquired	Printed	Enlargement	Yes	

Table 2: Results from Experiment 1

Hz								
Hz	1	2	3	4	8	16	32	
0	4.9	4.9	4.9	4.8	4.7	4.7	4.2	
1	-	4.2	4.4	4.4	4.7	4.8	4.7	
2	-	-	3.1	3.5	4.2	4.6	4.6	
3	-	-	-	2.4	3.3	3.9	4.1	
4	-	-	-	-	2.8	3.5	3.8	
8	-	-	-	-	-	1.9	3.1	
16	-	-	-	-	-	-	1.9	

3.2 Experiment 2 - Ability to track different frequencies

In Experiment 1, we identified the frequency at which discrimination was easy when the shape did not change. The purpose of Experiment 2 was to identify the frequency at which it was easy to distinguish when the pin moved continuously, as in an actual game.

3.2.1 Participants. Six males (aged 19 to 52 years, 28.8 ± 13.5) volunteered as participants in this study. Table 3 shows the details of the participants. Three of them were blind, and three were visually impaired.

3.2.2 Procedure. The participants ranked the ease of tracking the moving pins. As shown in Figure 4, the pins moved around the square, drawn as a set of 8×8 pins as in Experiment 1. Similar to the case in Experiment 1, these pins corresponded to 0, 1, 2, and 3 Hz from the upper left to upper right, and 4, 8, 16, and 32 Hz, from the lower left to lower right. The process was initiated from 0 Hz, progressed to 1 Hz, 2 Hz, and so on up to 32 Hz. The process was then initiated backwards, from 32 to 16 Hz, 8 Hz, and finally down to 0 Hz. The participants were not provided with the instructions on the operation.

3.2.3 Results. The order of the ease of tracking was 16, 32, 8, 0, 4, 3, 2, and 1 Hz. The participants commented that the pins

**Figure 4: Setup for Experiment 2.**

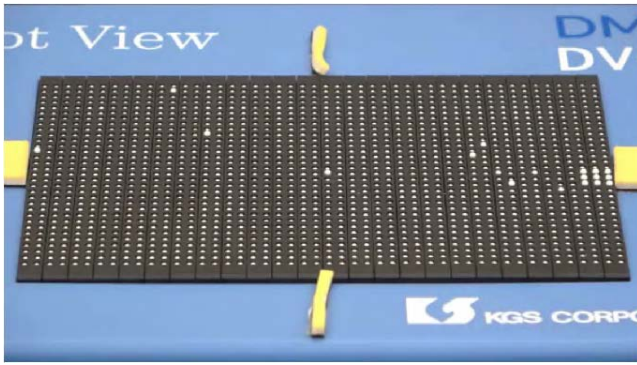
seemed to disappear at lower frequencies, and that it was difficult to discriminate between 0 and 32 Hz.

3.3 Experiment 3 – Understanding the formation of the team

Finally, based on the results of Experiments 1 and 2, the player positions as extracted from the actual games were displayed on the graphic display; further, it was verified as to whether the participants could distinguish between the two teams, designated by different frequencies.

Table 3: Participants in Experiment 2

	Age	Visual acuity (yard-pound system)	Congenital/acquired	Type of characters used	Method to obtain information	Experience of habitually watching (hearing) sports	Remarks
1	19	under 20/2000	Acquired	Braille	Voice, braille	Yes	
2	19	under 20/2000	Congenital	Braille	Voice, braille	Yes	
3	20	20/1000 – 20/666	Congenital	Braille, printed	Voice, enlargement, braille	Yes	
4	20	20/1000 – 20/666	Congenital	Printed	Same as sighted person	Yes	Blind soccer player
5	52	under 20/2000	Congenital	Braille	Voice, braille	Yes	
6	43	under 20/2000	Acquired	Braille	Voice, braille	Yes	

**Figure 5: Experimental setup for Experiment 3.**

3.3.1 Participants. The participants were the same as those in Experiment 1.

3.3.2 Procedure. As shown in Figure 5, the participants verified whether they could distinguish the members of the two teams in the actual game at different frequencies. They also verified whether they could understand the formation of each team, which changed dynamically over time. For the actual game scene, a scene of 7 s was employed until one team scored the goal. Based on the results of Experiments 1 and 2, the scoring team was set to 32 Hz, the other team was set to 8 Hz, and the ball was set to 0 Hz. Each player displayed with 1 pin and the ball was displayed with 3×3 pins. The participants were not provided instructions on the operation.

3.3.3 Results. From the experiment, six of the seven participants were able to identify each team member. Additionally, one of the seven participants was able to understand the formation. Several participants commented that they could distinguish the detailed movements of the players, which was not possible with the live announcements.

4 DISCUSSION

The results of Experiment 1 show that 0, 1, 2, 8, and 32 Hz were suitable for distinguishing two different vibrations. The results of Experiment 2 indicate that the order of the ease of tracking the

moving pins was 16, 32, 8, 0, 4, 3, 2, and 1 Hz. Based on both the results, one team was set to 32 Hz, the other team was set to 8 Hz, and the ball was set to 0 Hz in Experiment 3. Consequently, most participants were able to identify each team member.

This system can be displayed simultaneously on four tactile displays to ensure that four field players are displayed simultaneously. Therefore, it is applicable not only for spectators, but also for players as an effective method of strategy. However, the position of the camera must be set high enough to capture the entire field at once, and positioning the camera on a tall pole might be dangerous and difficult to execute in a local game. It is therefore necessary to identify workarounds to address this issue in future studies.

Kinesthesia is as important as the visual and auditory senses for safety, comfort, and quality of entertainment. However, the presentation methods for kinesthesia have not been developed as much as those for visual and auditory senses. This is because the mechanisms of kinesthesia have not been clarified as much as those of the visual and auditory senses have. We will continue to develop methods to present kinesthetic sensations by considering tactile sensations.

5 CONCLUSION

To develop a blind football play-by-play system for visually impaired spectators, we have herein developed an experimental system that detects the positions of the players and ball on a field, using image processing technology and deep learning, and that displays their positions on a refreshable tactile display. The results establish the possibility of displaying arbitrary positions of the players and ball on the tactile display. In the future, we plan to add a synthesized voice to the output method.

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